Gawtam Chithra Ramesh

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Education

KTH Royal Institute of Technology — Aug 2024 – Jun 2026
Master of Science in Systems, Controls, and Robotics - 4.62/5.0 — *Stockholm, Sweden*

Indian Institute of Technology Madras — Aug 2019 – Jun 2024
Dual Degree (B.Tech in Engineering Design and M.Tech in Robotics) — *Chennai, India*

- Awarded (among 30 of 800+) the IIT Madras **Young Research Fellowship** based on remarkable research potential.
- Secured **97.64** Percentile in JEE Advanced and **99.10** Percentile in JEE Mains among **1.5M** candidates.
- Secured an All India Rank of 52 out of 50k in Undergraduate Common Entrance Examination for Design.

Skills : Python, C++, C#, MATLAB, ROS2, Unity3D, Fusion360, MuJoCo, Pytorch, OpenCV, YOLO, Docker, Git

Professional Experience

Guiding Generative Flow with Formal Specifications — (Ongoing) Jan 2026 - Jun 2026
ABB Robotics — *Västerås, Sweden*

- Using generative flow matching for safe, compositional translation from natural language to
- control policies under Signal Temporal Logic constraints for long horizon industrial manipulation.
- Exploring reinforcement learning and geometric generative modeling for .

Device Research Intern on 3D Scene Graph — Jul 2025 - Dec 2025
Ericsson Research — *Lund, Sweden*

- Developed a robust 3DSG pipeline by integrating instance segmentation and tracking into a SOTA framework (HYDRA)
- Designed a synchronized RGB-D inpainting pipeline to remove privacy-sensitive objects from 3DSG in realtime.
- Benchmarked SOTA inpainting algorithms to balance reconstruction fidelity with real-time performance.

Graduate Robotics Intern on Modular Collaborative Robots — Dec 2022 - Jul 2023
Systemantics India Pvt. Ltd. — *Bangalore, India*

- Developed backward-chained Behavior Trees for 6DOF manipulator to enable robust, long-horizon trajectory planning
- ROS2
- actuation via low-latency communication with motor controllers using D-Bus and SocketCAN.

Research Experience

Taxonomy-guided VLMs for Deformable Object Manipulation — Jan 2025 - Nov 2025
Prof. Florian T. Pokorny, Deptartment of Computer Science, KTH — *Stockholm, Sweden*

- Integrated taxonomy-guided vision-language models (Gemini, Qwen) to interpret DOM scenes and map motion primitives
- Performed quantitative analysis to identify failure modes and guide iterative improvements to the taxonomy and prompts
- Currently integrating multi-modal data and temporal cues to enhance the perception pipeline's robustness.

Low cost autonomous ultrasound using SfM — Jan 2024 - May 2024
Prof. Nirav Partel, Department of Engineering Design, IIT Madras — *Chennai, India*

- Developed an autonomous ultrasound scanning system using ROS, integrating a 5-DoF robotic arm with a single RGB
- Single RGB Camera to achieve cost-effective 3D perception for medical imaging.
- Integrated perception algorithms within ROS, processing 3D point clouds from RGB images t

Technical Projects

Robot Learning and Embodied AI | DD2601 - KTH — Sep 2025 - Oct 2025

- Implemented a 3D semantic object mapping pipeline using RGB-D data from ARKitScenes, integrating OwlV2
- for open-vocabulary object detection, SAM for segmentation, and CLIP for semantic grounding.
- Built a 3D semantic mapping pipeline using RGB-D data and foundation models

Deep Learning Advanced | DD2601 - KTH — Sep 2025 - Oct 2025

- Mean Flow
- SimCLR
- VAE

RL and MAS | DD2601 - KTH — Sep 2025 - Oct 2025

- MAPF
- Pacman RL - BT
- Minimax

Lead Author Publications

- Robust Object-layer Construction of 3DSG Using Instance Segmentation (IMPROVE26)(**Best Paper Award**)
- Synchronized RGB-D Inpainting for Privacy-Aware 3D Scene Graph Construction.(**ICPR26**).(Under Review)
- Scene Understanding in Deformable Object Manipulation via Taxonomy-Guided VLMs.(**IROS25** Workshop)